

Calculus III and the Mathematics of Insect Flight

Mathematics primarily concerns itself with its own terminology and laws, but it is integral to our understanding of physics. New applications of previously theoretical mathematical concepts continue to arise, especially in scientific fields. Biological systems are complex and cross-disciplinary, but physics and applied mathematics are often hidden inside real biological systems, such as animal motion. Insect flight is incredibly fascinating, especially the advanced aerial locomotion of the order Diptera, commonly known as the true flies. Flies have some of the greatest aerial mobility among all animals, being able to hover, turn, and maneuver in ways that look simple but are very physically complicated. I chose this topic for my paper because I am passionate about the intersection of my major, Computer Science, and wildlife, ecology, and biological research. The ways in which insect flight connects to Calculus III include vector-valued functions, acceleration, vector fields, and forces acting from multiple directions, such as inertia, gravity, and propulsion. Essentially, insect flight is the perfect example of Calculus III concepts used to describe real motion and airflow in living organisms.

Insect flight is more complicated than it may appear at first and is extensively studied by biologists, physicists, and engineers alike. “The flight of insects has fascinated physicists and biologists for more than a century” (Sane, 2003, p. 4191). Some of the older, simpler airplane concepts were not enough to explain flapping flight because insect wings do not move in a linear way but as a combination of multiple vectors, rotation, and interaction with moving air. New tools such as high-speed video, mechanical wing models, and computational models make insect flight easier to study than ever before. These tools often depend on measuring position, direction, speed, acceleration, and force.

Calculus III uses vector-valued functions like $\mathbf{r}(t)$ to describe the position of something moving through 3D space. A fly's body position, or even a single point on the surface of a fly's wing, can be thought of as a position vector changing over time. The derivative $\mathbf{r}'(t)$ gives velocity, meaning the direction and speed at that specific moment, just as its second derivative gives acceleration. Flies can perform very complex aerial maneuvers, such as flipping upside down midair and landing on the ceiling. This involves rapid acceleration in order to change upward motion and body orientation in a fraction of a second. A successful inverted landing involves "an initial upward acceleration followed by rapid body rotation and leg extension" (Liu et al., 2019).

Vectors have not only direction but also magnitude, and this is true for forces such as lift, drag, gravity, and propulsion. Flies must produce enough upward force to stay in the air while also controlling rotational motion for steering and complex movements forward, backward, and sideways. A fly's wings do not simply push air downward but rotate and interact with air left behind by previous wingbeats. Insect flight depends on "delayed stall, rotational circulation, and wake capture" (Dickinson et al., 1999, p. 1954). The wing can maintain useful airflow and lift during part of the wing stroke instead of immediately losing lift like a simple fixed wing, such as an airplane wing, might. When the wing turns around at the end of a stroke, that rotation and the remaining moving air can help produce extra force. In Calculus III terms, these forces can be described using vectors, while a dot product could measure how much of a force acts in a specific direction, such as upward lift or forward motion.

The rotation of the fly's wings is the key to rapid angular change during flight and redirects forces quickly. Insects can "modulate the direction and magnitude of flight forces" (Dickinson et al., 1999, p. 1954). When a fly changes the angle and motion of its wings, it

changes the aerodynamic force vectors acting on its body. If the net force changes, the fly's acceleration changes. This connects to the idea of a space curve in Calculus III, where the position of a moving point can be described by a vector-valued function over time. A fly's motion through the air can be thought of as a point tracing a curved path through three-dimensional space, but the actual motion is more complicated because that path is constantly affected by several changing force vectors, including lift, drag, gravity, and propulsion. In a simulation of fly movement, a computer model would need to account for position, velocity, acceleration, wing angle, and changing force vectors over time.

A vector field is a useful way to imagine the medium the fly is pushing and moving through: the air. In a vector field, each point in space has a vector showing the direction as well as the speed of flow at that location. A fly's wing creates complicated airflow because of the redistribution of the air and the altering of the vectors of existing particles. A leading-edge vortex "greatly enhances the forces generated by the wing" (Sane, 2003, p. 4191). A vortex is a spinning airflow that can help increase the aerodynamic force produced by the wing. This connects to Calculus III because vector fields describe changing flow through space, while flux describes how much flow passes across a surface. Full fluid dynamics is more advanced than what I have learned so far, but Calculus III provides the basic language for describing direction, flow, and force.

In our ordinary lives, flies are more often seen as an annoyance and are not really given more than a glancing thought, but flies are a vital part of the ecosystem as pollinators, decomposers, and prey animals. Their flight affects how they find food, escape predators, locate mates, and spread through habitats. A fly's ability to maneuver itself is important to understanding its biology and also to understanding how flies can have such a large effect on the

ecosystem. Outside of biological fields, a fly's movement is useful for engineering as it can be used to inspire new and more complex models of flight for technologies like drones and other aerial devices. One direct example of this kind of biological inspiration is the Harvard RoboBee project. Researchers created an insect-scale flying robot modeled partly after fly biology, with small flapping wings and a control system designed to manage the instability of flight at such a small size (Ma et al., 2013). Possible applications for this kind of work include environmental monitoring, search-and-rescue, and even crop pollination, which connects insect flight research back to both ecology and technology. As a Computer Science student with an interest in biology and entomology, this topic shows how math can support models, simulations, and animal-related technology.

This topic helped me personally connect how Calculus concepts are more than just abstract formulas and have real-life applications to physics and living movement. Vector-valued functions, velocity, and acceleration come together to describe something that seems exceedingly complicated when viewed as a whole. Mathematics allows complex problems to be studied quantitatively, which is especially useful when studying animal motion. This paper helped me personally see vectors as actual descriptors of real-life forces instead of only as arrows in a homework problem. This topic relates to my future interest in Computer Science, such as ecological data and simulation software, and I believe it prepares me to think more critically about the application of mathematics.

Fly flight is a real-world example of Calculus III concepts being applied. Vector-valued functions describe the position of the fly itself, derivatives describe the fly's velocity and acceleration, vectors describe the individual acting forces on the fly, and vector fields relate to the airflow around the organism. Both physical and mathematical models are used to understand

the movements of the animal, its hovering, wing rotation, and even complex movements like an inverted landing. Studying insect flight patterns and their relation to mathematics makes math feel less like disconnected formulas and more like a formal and quantitative language for understanding movement in nature.

References:

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